

Solve Multiplication and Division Equations

Lesson 7-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

*Multiplication and division are inverses: $3 \times 7 = 21$,
so $21 \div 3 = 7$*

Inverse operation

Write the definition:



$$3 \times 2 = 6$$

$3 \times 7 = 21$ means 3 groups of 7 equals 21

Multiply

Write the definition:



$$6 \div 3 = 2$$

*$36 \div 4 = 9$ means 36 split into 4 equal groups gives
9 per group*

Divide

Write the definition:

$$x + 2 = 7$$

balanced with =

$3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$ (x is alone)

Isolate

Write the definition:

$$4x$$

coefficient = 4

In $6x = 18$, the coefficient is 6; divide both sides by 6 to get $x = 3$.

Coefficient

Write the definition:

$$x + 2 = 7$$

balanced with =

$x = 3$ is the solution of $6x = 18$ because $6 \times 3 = 18$.

Solution

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can solve one-step multiplication and division equations using inverse operations.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inverse operation

Multiply

Divide

Isolate

Coefficient

Solution

1 An operation that undoes another, like division undoing multiplication, is an

_____.

2 To undo dividing by a number, I _____ both sides by that number.

3 To undo multiplying by a number, I _____ both sides by that number.

4 To get the variable alone on one side is to _____ it.

5 The number multiplied by a variable is the _____.

6 The value of the variable that makes an equation true is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $6x = 42$

- A. $x = 7$
- B. $x = 48$
- C. $x = 36$
- D. $x = 252$

Step 1 Divide both sides by 6: $x = 42 \div 6 = 7$.

Step 2 Check: $6 \times 7 = 42$ ✓

✓ **Answer:** A. $x = 7$

 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $n \div 3 = 8$

- A. $n = 24$
- B. $n = 11$
- C. $n = 5$
- D. $n = 3$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $9m = 63$

- A. $m = 7$
- B. $m = 54$
- C. $m = 72$
- D. $m = 567$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Vault B is sealed with $x / 3 = 6$. Crack the lock: what is x ?

A. $x = 18$

B. $x = 2$

C. $x = 9$

D. $x = 3$

Show your work:

2. Vault C is sealed with $5n = 45$. Crack the lock: what is n ?

A. $n = 9$

B. $n = 225$

C. $n = 40$

D. $n = 50$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

Sentence starter: Equation: _____. Solving: _____. The answer _____ is reasonable because _____. A division situation with the same answer: _____.

Show your work:

Reflect — Exit Ticket

Solve: $w / 8 = 6$

- A. $w = 48$
- B. $w = 14$
- C. $w = 0.75$
- D. $w = 2$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Inverse operation
2. **Notes 2:** Multiply
3. **Notes 3:** Divide
4. **Notes 4:** Isolate
5. **Notes 5:** Coefficient
6. **Notes 6:** Solution
7. **We Try (notes):** A. $n = 24$ — *Multiply both sides by 3: $n = 8 \times 3 = 24$. Check: $24 / 3 = 8$ ✓*
8. **You Try (notes):** A. $m = 7$ — *Divide both sides by 9: $m = 63 \div 9 = 7$. Check: $9 \times 7 = 63$ ✓*
9. **Try It 1:** A. $x = 18$ — *x is divided by 3, so multiply both sides by 3: $x = 6 \times 3 = 18$. Check: $18 / 3 = 6$.*
10. **Try It 2:** A. $n = 9$ — *The 5 is multiplying n , so divide both sides by 5: $n = 45 / 5 = 9$. Check: $5 \times 9 = 45$.*
11. **Exit Ticket:** A. $w = 48$ — *Multiply both sides by 8: $w = 6 \times 8 = 48$. Check: $48 / 8 = 6$ ✓*